Charts and diagrams — Tables · Averages · Pie charts · Stem and leaf · Scatter graphs

A calculator is allowed for CALC questions. Squared paper and a protractor are needed for the drawing questions. ★ = challenge question.

1 A pie chart shows 72 birds seen in a garden one day. The sector angles are: starling $150°$, pigeon $90°$, goldfinch $45°$, sparrow $45°$, other $30°$. How many of the birds were **a** starlings, **b** sparrows, **c** goldfinches?

> $360°$ represents 72 birds, so $1°$ is $72 \div 360 = 0.2$ birds. **a** $150 \times 0.2 = 30$ **b** $45 \times 0.2 = 9$ **c** 9. (Pigeon 18, other 6; total 72 ✓)

2 A survey asked people how many items they bought online in a week.

Items	0	1	2	3	4	5
Frequency	5	8	12	10	8	2

a Work out the range.

b Write down the mode.

c CALC Work out the mean, correct to 1 decimal place.

d Find the median.

> **a** $5 - 0 = 5$. **b** 2 (highest frequency, 12).
>
> **c** Total items $= 0 \times 5 + 1 \times 8 + 2 \times 12 + 3 \times 10 + 4 \times 8 + 5 \times 2 = 104$; people $= 45$. Mean $= 104 \div 45 = 2.3$.
>
> **d** 45 people → the 23rd value. Running totals: 5, 13, 25 — the 23rd is in the "2" group. Median $= 2$.

3 Draw a pie chart for the data in question 2.

> Angle $= \dfrac{\text{frequency}}{45} \times 360 = 8°$ per person: $0 \to 40°$, $1 \to 64°$, $2 \to 96°$, $3 \to 80°$, $4 \to 64°$, $5 \to 16°$ (sum $360°$ ✓). Label each sector.

4 The table shows what two families spent on food each week over a year.

	Mean	Median	Range
Petrou family	€85	€82.50	€38
Lam family	€75	€81	€24

a Write two sentences comparing the families' spending.

b Why is there unlikely to be a modal value for a family's weekly food bill?

> **a** On average the Petrou family spend more (mean €85 vs €75; medians €82.50 vs €81, so the typical week is similar — the Petrou mean is pulled up by some expensive weeks). The Petrou family's spending varies more (range €38 vs €24); the Lam family are more consistent.
>
> **b** Amounts like €83.47 are rarely exactly repeated, so no value occurs more often than the others.

5 Two pans of hot water were left to cool, one with a lid. Temperatures (°C) every 10 minutes:

Time (min)	0	10	20	30	40	50	60
Lid off	100	62	40	28	22	20	20
Lid on	100	78	58	42	30	24	20

a Draw both sets of data as line graphs on the same axes.

b What is room temperature? How do you know?

c Compare how long the pans took to reach room temperature.

d What was the difference in temperature after 30 minutes?

> **a** Time on the horizontal axis; two curves, both falling and levelling off.
>
> **b** $20\,^\circ$C — both graphs flatten out at that value. **c** Lid off: about 50 minutes. Lid on: about 60 minutes — the lid keeps the heat in, so it cools more slowly. **d** $42 - 28 = 14\,^\circ$C.

6 The masses (g) of ten eggs: 58.5, 61.3, 55.2, 58.6, 49.1, 45.2, 64.7, 61.2, 55.0, 59.5. Copy and complete the grouped frequency table with classes $45 \le m < 50$, $50 \le m < 55$, $55 \le m < 60$, $60 \le m < 65$.

> $45 \le m < 50$: 49.1, 45.2 → 2. $50 \le m < 55$: none → 0. $55 \le m < 60$: 58.5, 55.2, 58.6, 55.0, 59.5 → 5. $60 \le m < 65$: 61.3, 64.7, 61.2 → 3. Total 10 ✓.

7 Copy and complete the two-way table about people at a leisure centre.

	Tennis	Hockey	Badminton	Total
Female	20		15	67
Male		26		70
Total	44		35	

a How many people were playing hockey?

b How many females played hockey?

Female hockey $= 67 - 20 - 15 = 32$. Male tennis $= 44 - 20 = 24$. Male badminton $= 35 - 15 = 20$ (check: $24 + 26 + 20 = 70$ ✓). Hockey total $= 32 + 26 = 58$. Grand total $= 137$.

a 58 **b** 32.

8 A stem-and-leaf diagram shows reaction times (seconds): 10 | 5 7 · 11 | 0 4 9 · 12 | 6 6 8 9 · 13 | 2 5 5 6 8 · 14 | 1 3 3 (key: 10 | 7 means 10.7 s).

a What is the range?

b What is the median?

c What is the mode?

17 values in order. **a** $14.3 - 10.5 = 3.8$ s. **b** The 9th value: 10.5, 10.7, 11.0, 11.4, 11.9, 12.6, 12.6, 12.8, **12.9** → median 12.9 s. **c** 12.6 and 13.5 both appear twice — two modes.

9 The table shows the wingspan and leg length of some falcons (cm).

Wingspan	102	120	117	112	132	111	126	118	121	107
Leg length	40	47	45	44	54	45	50	43	50	47

a Draw a scatter graph (wingspan across, leg length up).

b Describe the correlation.

c Draw a line of best fit and use it to estimate the leg length of a falcon with wingspan 124 cm.

b Positive correlation — larger wingspan goes with longer legs — fairly strong (points close to a line).

c A line of best fit through the middle of the points, roughly from $(102, 40)$ to $(132, 54)$: at 124 cm it gives about 50 cm (accept 49–51).

10 CALC Estimate the total weekly pocket money given to children in a country using these facts: 70% of 5–16 year olds get pocket money; the average is €6.50 a week; there are about 8 million children aged 5–16.

$0.70 \times 8\,000\,000 = 5\,600\,000$ children; Math input error — about €36 million a week. (An estimate: the "average" and the 70% are themselves rounded.)

Check: pie chart — angle per item = 360 ÷ total · mean = total ÷ count, median = middle of the ordered list, mode = most common · compare data with an average *and* a measure of spread.